

Facundo Bacigalupi

Computer Analyst

+598 096 366 633
Montevideo, Uruguay
bacigalupifacundo@gmail.com

LinkedIn: [facundobacigalupi](#)
GitHub: [FacundoBacigalupi](#)
Portfolio: [github.io/cv-web](#)

Computer Analyst and advanced Computer Engineering student at Universidad de la República. I have experience in backend and web software development and maintenance, mainly with Java, Spring, Angular, TypeScript, databases, and REST/SOAP services. I am interested in working on real systems, understanding existing codebases, and building clear, maintainable, and useful solutions.

SKILLS

Languages	Java, Python, JavaScript, TypeScript, SQL, C/C++
Backend	Spring Boot, Spring Framework, REST APIs, SOAP, Java EE, Servlets, JSP
Frontend	Angular, React, HTML, CSS, Tailwind CSS
Databases	PostgreSQL, MySQL, MongoDB, SQLite
Testing and QA	Selenium, Cucumber, JUnit, Postman, TDD
Tools	Git, Maven, Docker, Kafka, Tomcat
Other interests	Flutter, Unity, OpenGL, TensorFlow Lite
Communication	Native Spanish, Advanced English

PROJECTS AND EXPERIENCES

Software Developer GEOCOM Uruguay 07/2022 — 01/2024

Participated in backend and web software development, maintenance, and testing tasks. Worked on promotion subsystems for sales and finance applications, fixing bugs and adding features using Java 6 and SOAP services. Also implemented automated tests for point-of-sale systems using Java 8 and Cucumber, helping validate critical business flows. Additionally, participated in accounting and reporting projects using Spring Framework, MongoDB, Kafka, and REST services. Also performed frontend maintenance with Angular and TypeScript, and implemented automated tests with Selenium and Cucumber.

Databases University Project

Designed and implemented relational database solutions. Worked with entity-relationship modeling, relational algebra, SQL, functional dependencies, normalization, transactions, concurrency, and recovery. Developed queries and analyzed schema integrity, efficiency, and consistency.

Computer Networks University Project

Studied the theoretical and practical foundations of the TCP/IP model, including application, transport, network, and link layer protocols. Developed a lab project using Mininet, implementing a client-server application for traffic generation and routing/forwarding functionalities for an IP router within an emulated network.

Software Engineering Project University Project

Continued and extended a system previously developed by another team. Added new features and developed an associated mobile application from scratch. This project involved understanding an existing codebase, adapting to previous architectural decisions, designing new functionality, and building user-oriented solutions.

Programming Workshop University Project

Developed a Java application using Java SE 8 and Java EE. Implemented web functionality with Servlets, JSP, and Tomcat, along with an administrator interface developed in Swing. Worked under an incremental methodology, progressively adding new features.

C/C++ Projects University Project

Developed applications in C and C++ focused on data structures, algorithms, object-oriented programming, design patterns, and large-scale programming concepts.

Mi Stock Personal Project

Developed a mobile inventory application for Android using Flutter. The app allows users to register products, manage stock quantities, store photos, scan barcodes, and view basic stock movements. The project allowed me to practice mobile development, local persistence with SQLite, state management with Riverpod, UI design with Material 3, and internationalization. I also explored an experimental offline visual search feature using TensorFlow Lite, aimed at comparing locally stored product photos.

CipherChat**Personal Project**

Educational prototype of an end-to-end encrypted messaging application. Developed with Spring Boot, React, TypeScript, PostgreSQL, JWT, and WebSocket/STOMP. It implements authentication, direct chats, multiple devices per user, and message encryption per device, with keys generated in the browser.

HomeCinema**Personal Project**

Personal application for organizing and playing multimedia content on a local network. Backend developed with Spring Boot, frontend with React/TypeScript, and SQLite database. Includes local library scanning, playback progress tracking, TMDb integration, and video playback through HLS generated with FFmpeg.

Rubik's Cube 3D**Personal Project**

Developed a 3D Rubik's Cube game in Unity 6. Implemented interaction with a 3x3x3 cube, including sticker raycasting, layer rotations through click and drag, camera rotation, shuffle, and solved-state detection. Additionally, I explored an extension with reinforcement learning to train an experimental model aimed at neural search for cube movements.

EDUCATION

Computer Engineering, *Universidad de la República*

2018 - present